VINOTH RAJ R

AI Engineer | Software Engineer - AI Automation, LLM, CRM and Mobile

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SUMMARY

Software engineer with 3.5+ years shipping production systems, now building the AI layer of a financial-services sales operation end to end. At AssetPlus I own the Frappe CRM, replaced traditional tele-calling with a Sarvam AI voice agent and a root-free Flutter + Kotlin call-automation app, and run LLM QA on every recorded call. Python, Frappe/ERPNext, React/Next.js and LLMs (Sarvam AI, Gemini, OpenAI, Claude); open-source author of ai-knowhow and llm-json.

- **100% of new ad leads** get a first AI voice call, Sundays included, before a human dials.
- **92% to 100%** call-recording upload reliability with zero duplicates; app open at 1,000 calls cut from **19.1 s to 3.9 s**.
- **40% lower operational costs** and **50% higher team productivity** from 3 internal ERPNext apps.
- **6 marketing and business data sources** unified in one daily dashboard that management uses.

EXPERIENCE

Software Engineer, AssetPlus

Apr 2025 - Present

Chennai, Tamil Nadu | On-site | Financial services (mutual-fund distribution)

AI voice calling and LLM call QA

- Tele-callers spent most of the day dialing leads that went nowhere, night sign-ups waited, and screening was inconsistent. Designed an **AI voice-agent pipeline on Sarvam AI** with a rules engine in Frappe CRM: leads created 10 AM-7 PM are called immediately, night leads batch to 10 AM, failed calls retry after 2 days and never on Sunday. Every new lead now gets a first call and first-level screening, and hot leads reach the sales team's WhatsApp groups instantly.
- Management had no structured view of what happened on calls. Built a **server-side LLM QA pipeline**: Sarvam transcription, then four Gemini prompts (summary, tags, scoring metrics, QA questions) stored against the lead, with output normalisation, token and cost tracking, and a Gemini vs OpenAI benchmark harness. Every recorded call is now QA-scored, parse failures were removed, and models are chosen on evidence.

Call automation app (Flutter + Kotlin)

- No market product reliably captures SIM calls on ordinary Android phones, and the first version needed rooted custom-ROM devices that could not scale with the team. Re-engineered it into a **root-free app** (Shizuku, scrcpy audio capture) that logs every SIM and WhatsApp call and recording into the CRM. It runs on the phones agents already carry and, with the AI voice agent, replaced traditional tele-calling at significantly lower calling cost.
- Uploads died on bad networks and a race condition created duplicate CRM records. Built an offline-first queue, idempotent server upserts, hourly backfill and a fix for the concurrent-checkpoint race. Recording-upload reliability rose from **92% to 100% with zero duplicates**.
- The app slowed to 19.1 s to open at 1,000 calls. Moved recording lookup off the UI thread into one batched native index and switched to real-time mono 16 kHz recording. App open fell to **3.9 s** and recordings are about 6x smaller on disk.
- Battery savers could silently stop recording on a device. Added event-driven health reports (service up/down, missing audio, 2-hour heartbeat in working hours) with per-status de-duplication. Ops is alerted on WhatsApp the moment a device stops recording, and redundant status records fell from 13 to 0 per restart cycle.

CRM ownership and automation (Frappe CRM)

- Leads from Meta Instant Forms and the website needed one system for capture, prioritisation and follow-up. Set up and own **Frappe CRM end to end**: AI-driven lead prioritisation, Zoho sync, Google Calendar scheduling, WhatsApp and Slack alerts, and Frappe Insights dashboards. Sales and management now track team performance and lead conversion in one place.
- Meta ads optimised for form fills rather than real registrations. Integrated the **Meta Conversions API** so each CRM registration is sent back to Meta, and ad delivery now optimises for real outcomes.
- Hot leads needed a faster team response. Built WhatsApp automation (WPPConnect + Puppeteer) inside Frappe for instant hot-lead alerts and automated group onboarding, giving the team faster response on new leads.

Marketing and business metrics

- Marketing data lived in five places, each platform reported its own success, and nobody could say what a lead cost per campaign. Built the **Har Ghar MFD analytics dashboard** (React + Vite on Vercel): GA4 via Windsor.ai, Vimeo, Meta and Google Ads, Microsoft Clarity and CRM leads-to-deals, refreshed daily by GitHub Actions with a verification step, plus a Gemini **AI Insights tab** grounded on a bounded data brief. Management sees cost per real lead by campaign, platform-claimed vs CRM-received leads and AI-written executive summaries every day.
- The data showed 166 of 167 Google Ads leads reaching the CRM with no campaign, leaving **Rs 1.69 lakh of spend unattributable**. Traced it to the website dropping UTM and gclid parameters and specified the capture fix and a Frappe lead-and-deal API to close the attribution loop.

AI enablement

- The public site needed an AI assistant that was safe to expose and cheap to run; off-the-shelf widgets leaked API keys. Built a **Dify RAG assistant** with a Next.js server-side proxy, rate limiting, SSE streaming and two embed modes. It is designed for **1,000+ users a day** on free-tier hosting with no API key ever exposed to the browser.
- Departments needed a shared way to choose AI models for their work. Rolling out my open-source **ai-knowhow** skills company-wide so every department picks the right AI model and effort for the job and learns how AI-built work was done.

Senior Full Stack Engineer, Makto Technology

Dec 2022 - Mar 2025

Coimbatore, Tamil Nadu | Remote | UAE operations

- UAE operations needed internal tooling across CRM, fleet, healthcare and education. Led **5+ ERPNext implementations** and shipped 3 internal ERPNext apps that cut operational costs by **40%** and lifted team productivity by **50%**.
- UAE operations needed mobile ordering, payments and delivery tracking. Built a Flutter delivery platform (customer and delivery-agent apps with Firebase real-time data, Stripe payments, Google Maps live tracking) plus a React admin panel and warehouse management system, giving customers live tracking and the team role-based control of inventory and master data.
- Data collection was manual and deployments needed to be more reliable. Integrated LinkedIn and Google Forms with ERPNext and set up Git-based CI/CD, automating data collection and improving deployment reliability.

OPEN SOURCE

ai-knowhow | MIT | Agent Skills for Claude Code, claude.ai, Codex, ChatGPT | github.com/VINOTH-RAJ-R/ai-knowhow

- Choosing the right AI model, effort and cost for a task, and learning how AI-built work was done, had no standard method. **modelwise** turns a basic prompt into a top-practitioner plan, the right model and reasoning effort per phase, a cost comparison and a model-specific prompt, from a validated registry of 61 Anthropic and OpenAI models; **modelwise-run** executes it on the chosen Claude model. **retrace** turns a finished AI task into a saved lesson. Users get the right model at a known cost, with no server, API keys or data collection.

llm-json | MIT | Python 3.9+ | zero runtime dependencies | CI-tested | github.com/VINOTH-RAJ-R/llm-json

- LLMs return JSON wrapped in preambles, code fences, truncation or single quotes. Built staged, opt-in repairs that extract a usable object and, on failure, raise with the raw response, the stage reached and caller context. Production parse failures become debuggable instead of silent.

SKILLS

AI and LLMs: AI automation, LLM pipelines, Sarvam AI (voice agents, STT), Gemini, OpenAI, Claude, Dify / RAG, prompt engineering, agent skills, LLM benchmarking, structured-output parsing, token and cost tracking

Backend and CRM: Python, Frappe / ERPNext, Frappe CRM, REST APIs, webhooks, cron / Redis queues, MySQL, PostgreSQL, Zoho, Meta Conversions API

Mobile: Flutter (Dart), Kotlin, Android foreground services, telephony, Shizuku, Provider, Dio, offline-first sync

Frontend and Analytics: React, Next.js, TypeScript, Vite, GA4, Windsor.ai, Meta and Google Ads data, Microsoft Clarity, Vimeo API, Frappe Insights

Automation and DevOps: WhatsApp (WPPConnect, Puppeteer), Slack, Google Calendar API, Stripe, Google Maps, Firebase, Git, GitHub Actions, CI/CD, Vercel, Linux, Claude Code

Profiles: LeetCode | Stack Overflow | Frappe Community

EDUCATION

Bachelor of Information Technology, Sona College of Technology, Salem | Jun 2022 | CGPA 7.64